
Software Requirements Specification

for

**<Gorkha Online Supplement
Shop>**

Prepared by

19 Apr 2025

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1. Introduction

1.1 Purpose

The entire goal of developing the Gorkha Online Supplement Shop System is to build a responsive, secure and easy to use e-commerce platform where the customers can browse and purchase a variety of supplements related to health and fitness. Features that the system provides include product filtering, detailed descriptions, cart management, and real time, order tracking that are all designed to improve the online shopping experience.

1.2 Document Conventions

In order to guarantee readability and ease of reference for all parties engaged in the development of the Gorkha Online Supplement Shop System, this Software Requirements Specification (SRS) document adheres to clear and uniform formatting and standards.

- To identify distinct sections, all headers and subheadings are written in bold.
- Unless otherwise specified, high-level functional requirements inherit the priority status of the module or feature to which they are assigned.
- The term "System" describes the complete platform of Gorkha Online Supplement Shop.

1.3 Intended Audience and Reading Suggestions

- **Developers:** To comprehend the architecture, functionality, and specific implementation needs of the system.
- **Project Managers:** Project managers are responsible for monitoring the project's scope, progress, and alignment with its objectives.
- **Testers:** Create test cases according to both functional and non-functional specifications.
- **End users and administrators:** to gain a general understanding of the system's behavior and characteristics.
- **Documentation writers:** They are responsible for creating support materials and user guides depending on system features.

This SRS's remaining sections are structured to precisely outline the specifications and design standards for the Gorkha Online Supplement Shop System.

- **Section 1- Introduction:** The aim, scope, document conventions, target audience, and references are all covered in this section.
- **Section 2 - Overview:** A high-level overview of the system's operations, user roles, environment, limitations, and dependencies are given.
- **Section 3-Requirements for the External Interface:** Explains how users, hardware, software, and networks will all interact with the system.
- **Section 4 - System Features:** Outlines Specific Functional Requirements and Anticipated System Behavior.
- **Section 5 - Nonfunctional Requirements:** Performance, security, dependability, and quality of the system are covered in this section.

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2. Overall Description

2.1 Product Perspective

The Gorkha Fitness Supplement shop is a well-functioning e-commerce website which offers customers to buy fitness supplements online.

User Interface (Front end)

Customers can create their own personal account, log in to the website. This component includes filtering the product, searching the product and adding the product in the digital cart.

Admin side (Back end)

The admins can view sales reports, manage the users, and send promotional offers and other resources to the existing customers. Admins can login and manage all the data on the website which includes editing/adding/removing of products on the website.

Database system

A well-designed centralized database system for all the data such user's login details, inventory reports, product information. This will store data and allows editing and manipulating data as per requirement ensuring the user data security.

2.2 Product Functions

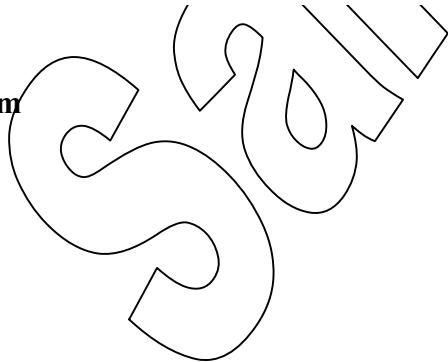
User side

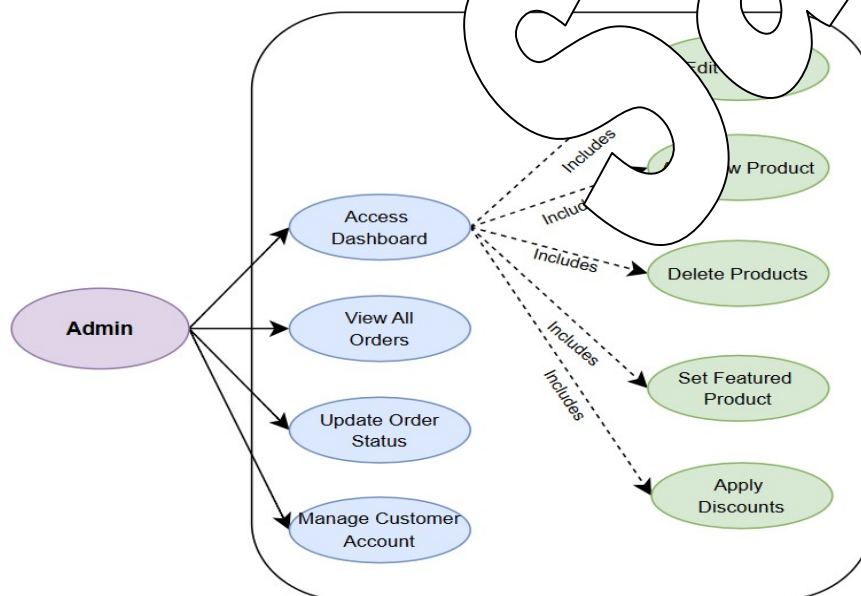
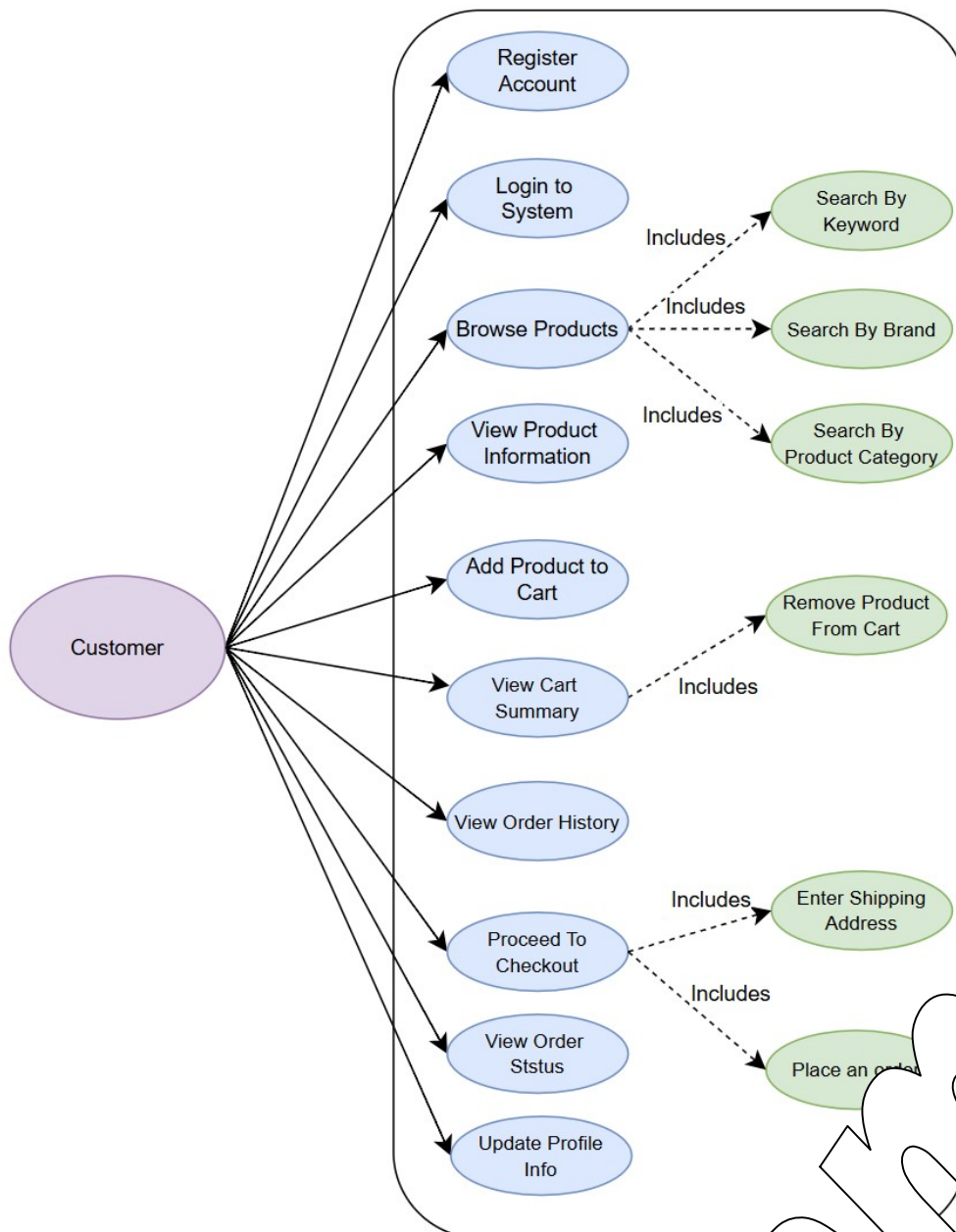
User side

FEATURES	DESCRIPTION
User Registration and login	The users can create their own profile, manage their user profile and create their own password.
Browsing the product	Users can browse different products, view different categories, filter the product according to the requirement.
Add product to cart	Users can add their selected product to their digital cart and make the payment as required.
Product details	See product specifications, image, price and availability of the product.

+ Admin side

Features	DESCRIPTION
Product management	Admins can add new products, remove product listing and make new changes to the product listing.
Inventory management	Monitoring the product stock level, marking the product as out of order if not available.
Discount coupons	Create discount offers, generate discounts coupons with codes and expiration date.

Use Case Diagram



2.3 User Classes and Characteristics

Gorkha fitness supplement shop can support multiple types of users with different requirements. All the users are divided into different classes.

Customers

These are the end users who browse through the ecommerce website and make the purchase. All customers have different levels of technical knowledge.

Admin

In this system admins have full access to the system and able to manage all the functions of the website. They are normally with good technical understanding and have the proper knowledge about how the system works.

Guest user

These are the users who are basically able to browse the website without creating their profile. They are not able to add product to the cart, make payments or leave user review.

User Class	Access Level	Technical knowledge	Use frequency	Importance
Customer	Basic	Low to medium	Medium to high	Very important
Admin	Full	High	High	Very important
Guest user	View only	low	Low to medium	Medium to high

2.4 Operating Environment

Gorkha fitness supplement shop is a ecommerce platform developed to be fully online, web-based application that this accessible form different devices.

Client side

The system will support various devices and operating systems providing a wide range of access for all the user, admins. Some supported devices are.

Desktops, laptops, smartphones, tablets.

Supported operating systems are:

Windows 10 and above, macOS, Android, iOS 12 and above.

As this being fully functional website, it is supported by different web browsers like Google Chrome, Mozilla Firefox, Safari.

Hosting and Backend infrastructure

The backend system can handle all the business logic and database operations. Some of the resources we are using for the backend technology are webserver: Apache HTTP, JavaScript for all the coding. HTML, CSS for the basic structure and frontend development. My SQL is used for all data related operations.

Third party integration

To enhance the third performance of the website there will be some third-party system included on the website. Different payment gateways like: Banks, PayPal.

2.5 Design and Implementation Constraints

For the development of the e commerce there will be different constraints that may limit and reduce the functioning ability of the final product.

Technical constraints

Advanced technologies like Nodejs and React may not be used due to lack of expertise. No real time processing like customer live chat and live support as this feature requires advanced infrastructure like web sockets and dedicated servers.

Budget constraints

As the project being student capstone, the team will use free or low-cost resources such as shared hosting platforms this limit the access to the premium features for the website.

Time constraints

As the time frame for the project is 12 weeks, which limits the time for developing and testing of the website so all the planned features may not be implemented.

Hardware constraints

The website will run on shared hosting servers, which will have limited processing power, storage which may impact the performance during the high traffic of data.

Software Limitations

Using simpler technologies like HTML, CSS, JavaScript instead of advanced framework may limit the performance in future use.

Security Constraints

Advanced features like multi factor authentication is not available in the initial stage.

Cooperate or regulatory policies.

The website must follow various ecommerce regulations which include consumer law, taxation, and refund policies which may require additional features to be developed. We should follow proper data protection law and procedures which will limit data storage platforms and data sharing platforms.

2.6 User Documentation

User documentation is the written or digital materials that provide the full information and guide the users to run and operate the website in the responsible way.

User Manual

This document includes the proper guide about how to use the website. It will include information about the different feature like login, browsing the product and many more. This document consists of information about all modules in sequential order so can be used easily as a requirement.

Administrative Guide

This guide is for the administrative group to manage the backend of the website. Information about different topic will be included in this section like how to add a new product in the product list, how to make changes in the product listing and offers.

Online Help Guide

This document will contain information like how to change passwords, how to solve payment issues and ways to report them and other common troubleshooting.

Documentation Standards and Format

All the text documents should be saved and stored as pdf format. And in HTML format for the web-based help to the user.

The language will be English, simple and less technical term for the east access to the user.

Different images and icon should be included in the document as required.

All the documents will be presented in a sequential order following the proper format like Title, heading, body.

2.7 Assumptions and Dependencies

As developing the ecommerce there can be several factors that may affect the result of the project. Some of the assumptions that may affect the results are mentioned below:

Assumptions Regarding development Tools

As we are using MySQL it is used for database management, depending on the nature of storage data if the database needs change there will be more changes required. As basic Resources like HTML, CSS AND JavaScript are being used it will more likely the feature and interface will be simpler, and many changes must be made in the future if the website has to extend big.

Assumptions regarding user experience

As the website is compatible with all the web browsers and devices but some with outdated versions may have compatible issues and may require extra work.

Assumption on Budget Timing

It is assumed that the website will be ready at the scheduled time but the delays on resources like developers are unavailable and other issues could delay the project. If the cost of technology, we will be using increases it will be hard to stay on the outlined budget.

3. External Interface Requirements

3.1 User Interfaces

A user-friendly interface system in the Gorkha Fitness Supplement Shop makes possible smooth administrator and customer interactions with the platform. Customers benefit from a current responsive design approach through this system so they can access content through multiple devices such as computers together with tablets and mobile phones.

3.1.1. Customer Interface:

Module	Description
Home Page	Shows discounted and featured products and latest reviews from the customers
Product Page	Dropdown menu for categories and product cards with ability to add to cart.
Cart & Checkout	For quantity increment or decrement, shipping address entry and order placement
Login or Register	Includes separate forms for new user registration and existing user login.

3.1.2 Admin interface:

Module	Description
Admin Dashboard	Shows a summary of products, sales customers in graphical and card format.
Product Management Panel	This panel allows admin to overall management of product I.e., adding removing, changing products type or description, and applying discount.
Order Management Page	Visualize list of orders and can change their status.
Customer Management Page	Shows all the registered customers and their previous orders in a list.
Sales Management	Can change the status of the order i.e., pending, completed or cancelled.
Logout Button	Helps admin to securely logout.

3.1.3. Standards and consistency

The website interface will and should follow UI/UX Standards to ensure:

- Consistent button placements (e.g., Add to Cart, Submit, Cancel).
- Clear labeling of form inputs and dropdowns.
- Minimal use of colors and graphics to maintain quick load times.
- Real-time feedback mechanisms (e.g., loaders, status updates).

These practices enhance the user experience and ensure higher satisfaction and engagement (Zhao et al., 2021).

3.2 Hardware Interfaces

The system operates with standard general-purpose clients and servers as illustrated in the following description.

3.2.1. Client-Side Hardware:

Devices

- Desktop Computers (Windows, macOS, Linux)
- Laptops
- Smartphones
- Tablets

Requirements: All these devices should support and run modern web browsers, i.e., (Chrome, Safari, Firefox, Opera) and should have a stable internet connection.

3.2.2. Server-Side Hardware

Hosting Server: This application can be deployed on in-premises hardware or in a cloud-based platform with the Specifications of:

- Processor: Quad-Core CPU (2.5 GHz or higher)
 - RAM: Minimum 8 GB
 - Storage: SSD with at least 100 GB for storing product images and customer data
 - Network: High-speed internet connection with 99.9% uptime
- Operating System: Linux-based servers (e.g., Ubuntu 22.04 LTS) are recommended for hosting the web server, database, and backend services.

3.2.3. Hardware Interface Characteristics

Communication between client devices and server occurs via standard networking protocols (HTTP/HTTPS).

The server may also interface with:

- External storage for backup and logs
- CDN (Content Delivery Network) services for faster image and content delivery (optional)

This physical layout enables a system which responds well while using resources efficiently and supports a high number of users simultaneously with administrative management functions that operate optimally.

3.3 Software Interfaces

The Gorkha Fitness Supplement Shop runs various software components which enable users to navigate through a convenient online shopping process. System interfaces establish operational rules between frontend and backend elements in addition to database components.

3.3.1. Frontend Technologies:

- HTML5, CSS3, JavaScript: This software is used for designing the client-side interface, making responsive and accessible user interaction.
- Bootstrap (v5.x): It is a framework used to make sure the site is mobile friendly and looks identical in all browsers.

3.3.2. Backend Technologies:

The backend system will be developed using PHP which will manage user login, order processing, registration, product management, and all the backend related tasks and will receive and send data from the front end in JSON format.

3.3.3. Database:

We will be using MySQL to store all the data. And connected to PHP backend. i.e.,

- User information.
- Product Details
- Cart items.
- Order history.

3.3.3. Other Details:

The system will use sessions to keep customers logged in, will validate data to prevent against any incorrect input or misuse and the entire system will work on a LAMP stack (Linux, Apache, MySQL, PHP)

3.4 Communications Interfaces

Gorkha Fitness Supplement shop will use basic internet communication methods to allow customers and the system to interconnect and interact safely and efficiently.

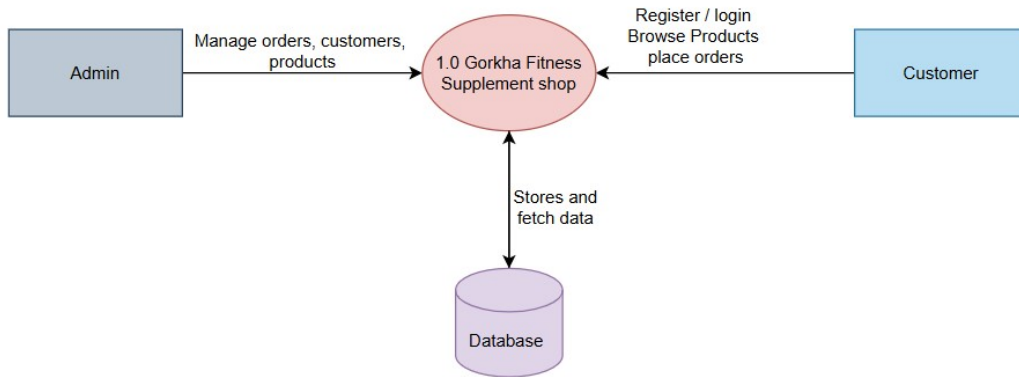
- Client-Server Communication: All communication between customer's web browser and web server will be passed through HTTP and HTTPS Protocols.
- Data Exchange: Frontend and backend system will use JASON format using AJAX to send and receive data and this allow smooth actions like adding to cart or searching products.
- Email Communication: The system will use SMTP for sending confirmation emails to customers (Password reset, Order Confirmation)
- Security: All the confidential information will be encrypted using SSL/TLS and the admin panel is protected with secure login to prevent any unauthorized access.

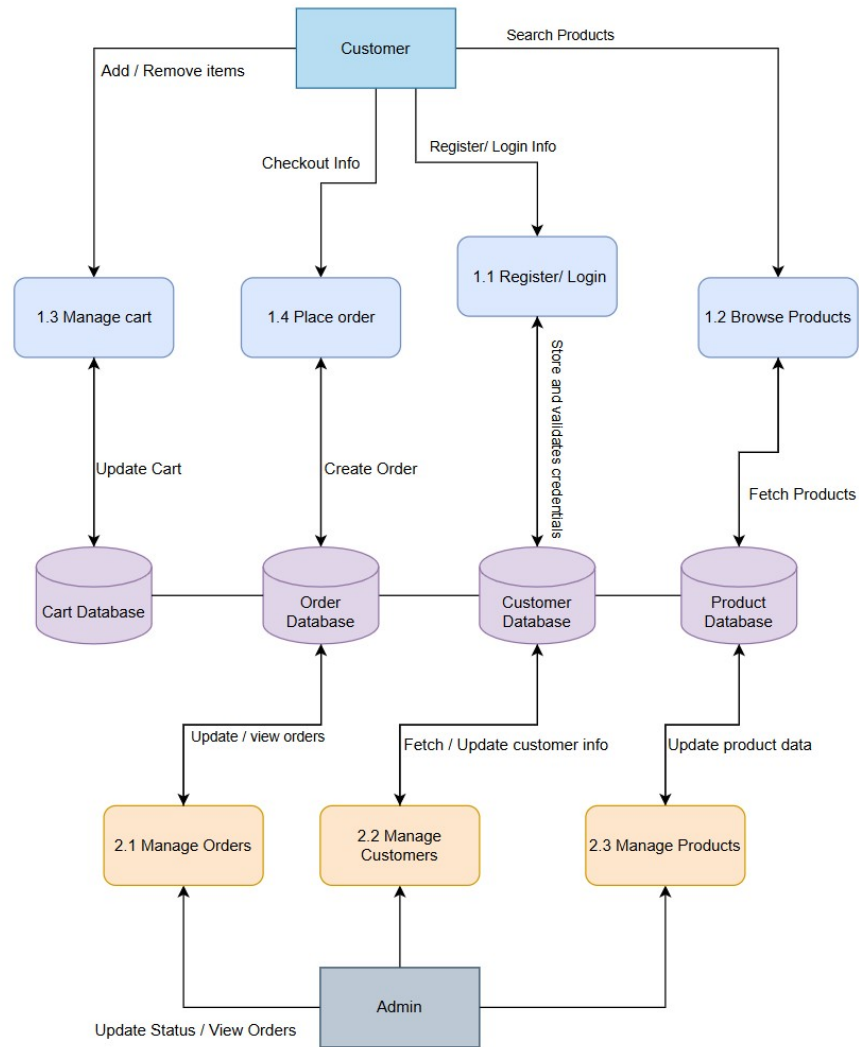
3.5. Data Flow Diagrams

3.5.1. Level 0 DFD

3.5.2. Level 1

DFD





4. System Features

4.1 User Registration And login

4.1.1. Description and Priority

This feature allows the user to create accounts on the platform.

Priority: High

4.1.2. Stimulus/Response sequence.

- Users click on the login page.
- Users enter the login credentials (login id, password, Gmail Id).
- System verifies the information.
- If the credentials are valid, it grants access to the system.
- If not valid declined the system access.

4.1.3. Functional Requirements

- REQST- 1.1: The system allows the user to register to the website using email and password.
- REQST-1.2: The system validates the uniqueness of id, password.
- REQST-1.3: The system saves the details in the database.
- REQST-1.4: The system allows users to log in using valid credentials.
- REQST-1.5: Provide error message if the credentials are not correct.
- REQST-1.6: Allow access if the credentials are correct.

4.2. Product Browsing

4.2.1. Description and priority

This feature allows the user to browse different fitness products according to their category and brand.

Priority: high

4.2.2 Stimulus/Response sequence.

- Users select the product according to category.
- The system filters the product according to the selected category.
- Users can sort the selected product by price range, name.
- Product listings are updated automatically.

4.2.3. Functional Requirements

- REQST-2.1: The system displays all the available products along with their name, images, category, price.
- REQST-2.2: System support filtering based on category, brand, price.'
- REQST-2.3: System allows shorting the product based on price and ratings.
- REQST-2.4: The system loads the product data dynamically from the backend.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Gorkha Online Supplement Shop System offers a smooth purchasing experience for customers and effective administration features for administrators. It is made to be extremely responsive, safe, and easy to use.

- Works perfectly During times of strong traffic, such as promotions or the debut of new products, this guarantees scalability.
- To ensure seamless interaction and user satisfaction (such as order status updates and admin-side issue messages) must appear on the user interface within three seconds.
- To guarantee quick navigation and reduce user annoyance, all important user pages, including the homepage, product listings, and product details, should load in less than two seconds under typical internet conditions.

5.2 Safety Requirements

Appropriate security measures must be in place to preserve system dependability and user confidence because the platform manages inventories, sensitive user data, and financial transactions. To protect users, uphold legal compliance, and stop abuse, certain measures must be put in place.

- **System and Data Protection:** To prevent system crashes or data corruption, the software must have error-handling techniques, access control, and input validation.
- **Preventing Unauthorized Access:** To avoid unauthorized modifications to inventory or customer orders, admin functionalities must only be available through secure login authentication.
- **Regular Backups:** The system should perform routine backups to preserve data from accidental loss or hardware failures.
- **Transaction Safety:** Systems, such as confirmation prompts, should be in place to stop duplicate orders or unsuccessful payments.

5.3 Security Requirements

Ensuring client trust and safeguarding sensitive data handled by the Gorkha Online Supplement Shop System require security and privacy. In addition to guaranteeing the safe management of data including login passwords, personal information, and order history, the platform must secure user and admin interactions.

- **User authentication:** Secure login credentials are required for both admin and consumer users.
- **Access Control:** Only authorized users should be able to access administrative features like order tracking and product management. It is necessary to enforce role-based access.
- **Data Transmission:** To avoid interception or manipulation, all data sent between users and the server must be encrypted using HTTPS.
- **Privacy Trust Certification:** Obtaining Privacy Trust Certification facilitates the development of open and honest customer interactions. When a website displays the Privacy Trust seal, users may feel secure knowing that no third parties will see their personal information.

5.4 Software Quality Attributes

In order to provide a safe, dependable, and responsive user experience, the Gorkha Online Supplement Shop System was built. The following qualities are thought to be crucial for fulfilling development standards and user expectations:

- **Usability:** All users, including new users, should be able to easily navigate the system. It should be easy to use features like product search, filtering, and checkout.
- **Reliability:** During routine tasks like adding products to the basket, processing orders, or updating product listings, the platform must operate reliably and without errors or crashes.
- **Availability:** To allow customers to shop whenever they choose, particularly during sales or periods of high traffic, the system must be available around-the-clock.
- **Scalability and Adaptability:** Without requiring significant adjustments to the system design, the system should be able to accommodate future expansion, such as the addition of additional product categories or the management of higher user traffic.
- **Security** Without jeopardizing data or functioning, the platform must manage large inputs, illegal access attempts, and unexpected system actions.

5.5 Business Rules

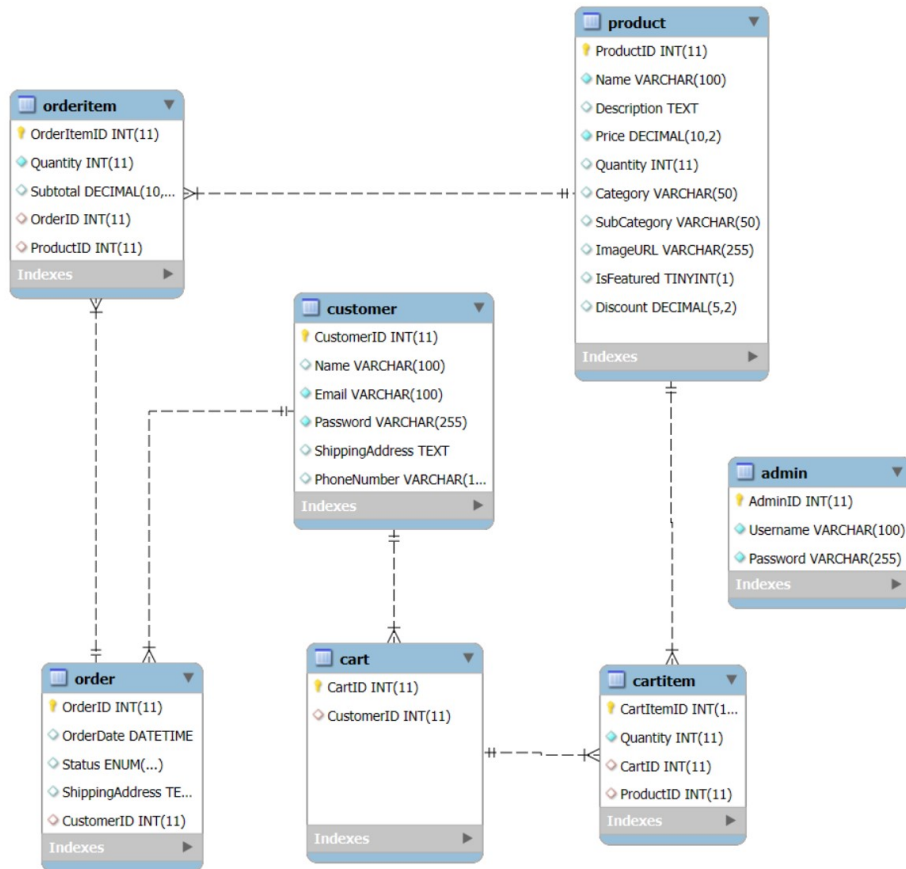
Gorkha fitness supplement shop proceeds some of operational principles for the website some are mentioned.

Operating principles	Implications
Only valid user and registered user can place the order on the platform.	User registration process before checkout process.
Administrators can add, edit the product from the catalogue.	Admin dashboard with product management features.
Only verified payment methods are accepted at checkout.	Integration with secured and verified payment gateways.
Users must agree the term and conditions before completing the purchase.	Check box for agreement document during checkout process.
Discounts code and coupons can be used by valid user in valid time.	Coupon management system.

6. Other Requirements

Appendix A: Glossary

ERD (Entity Relation Diagram)



SOL

Individual Contribution Statement Form**CONFIDENTIAL**

Note: To be completed individually on completion of the project. A signed hard-copy to be submitted to your Project Lecturer and a soft copy to be uploaded to the Moodle link. 35

Team Name: _TEAM 4_____

Lecturer Name: ___MEHEDI HASAN_____

(Please respond to the relevant box below)

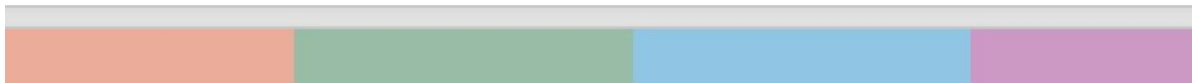
☒ All work has been equally and fairly distributed and all team members have contributed more or less equally. I think it is fair that we all receive the same team mark for all tasks completed by the team. (Write brief notes in the box below:-

☐ Work has **NOT** been equally distributed. I would suggest that the individual contribution for each task as listed could be summarised as follows:

Tasks	Team member Name	Individual Contribution % (Score out of 100 for each task)	Comments
	Member 1		
	Member 2		
	Member 3		
	Member 4		
	Member 5		

Note: Type in, then print and sign.

Additional comments/notes- Add here (use additional pages if required)



Signed: ...RANJIT.....=

Date: ...20 APRIL.....



STUDENT GROUP WORK GUIDELINE

Unit Title:	CAPSTONE PROJECT
Assessment Item Title	SRS REPORT
Assessment Due Date:	20 APRIL
Group Members	1. RANJIT KHANAL
	2. KRISHNA CHAULAGAIN
	3. PRINCE THAPA MAGAR
	4. DEEPAK PATEL
	5. ARJUN ADHIKARI

#1 Student Name:	RANJIT KHANAL
Tasks Responsible for:	1. PRODUCT PRESPECTIVE
	2. PRODUCT FUNCTION
	3. OPERATING ENVIRONMENT
	4.
	5.

#2 Student Name:	KRISHNA CHAULAGAIN
Tasks Responsible for:	1. USE CASE DIAGRAM
	2. DFD
	3. USER INTERFACE
	4.
	5.



#3 Student Name:	PRINCE THAPA MAGAR
Tasks Responsible for:	1. INTRODUCTION
	2. PERFORMANCE REQUIREMENT
	3. DOCUMENT CONVENTIONS
	4.
	5.





#4 Student Name: DEEPAK PATEL AND ARJUN ADHIKARI



#4 Student Name: DEEPAK PATEL AND ARJUN ADHIKARI

Tasks Responsible for: 1. SYSTEM FEATURES

Kent Institute Australia Pty. Ltd.

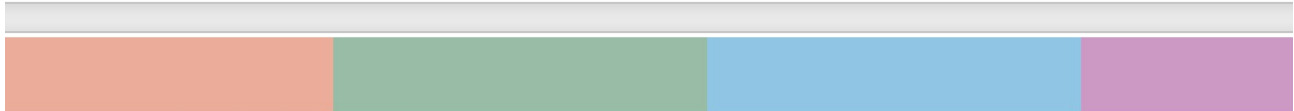
Student Group Work Guidelines

ABN 49 003 577 302 CRICOS Code: 00161E RTO Code: 90458

Version 2: 21st June, 2019

Page 2 of 4

TEQSA Provider Number: PRV12051



	2. BUSINESS RULES
	3. USER DOCUMENTATIONS
	4. DFD
	5.

Sample